Fine-Tune, Sanitize, Extract

Evaluating PII Sanitization and Training-Data Extraction

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Assignment summary. This report describes the implementation and results for Assignment 1. Part A compares the six specified LoRA and QLoRA configurations for masking personally identifiable information (PII), with A4 selected using validation scores. Part B evaluates A4 against regex and Presidio baselines: its overlap-based entity F_1 is 0.9339 on the in-distribution test set and 0.1958 on the out-of-distribution set. Part C applies the four required extraction attacks to an unmodified model and models trained on raw and sanitized text. Sanitization lowers targeted recovery from 7.5% to 1.5% of member prompts over ten sampled attempts, but increases held-out perplexity by 60.82%. These results show lower recovery in the targeted attack, poorer next-token prediction, and incomplete protection on unfamiliar data.

1 Experimental setting

The sanitizer learns to replace PII with labeled masks. The supplied splits are used in their original order [1]: 20,000 training rows and 2,000 validation rows for Qwen2.5-1.5B-Instruct. All six runs use one epoch, seed 42, AdamW with learning rate 2×10^{-4}, 3% warmup followed by linear decay, adapter dropout .05, and effective batch 16 (four batches of four). Loss applies only to the masked-text completion and its end-of-sequence token, not the prompt. The test sets are not used to choose the model.

2 Part A: adapter efficiency and model selection

Table 1: The six required sanitizer configurations. r is adapter rank and α its scaling parameter; memory is peak allocated GiB. Throughput counts input tokens/s. F_1 and leakage are validation rates; A4 is selected.

Run	Placement	r/α	Trainable%	GiB	Train (s)	Tok/s	IoU F_1	Leak
A1	q, v	8/16	0.071	3.598	5,865	1,314.0	0.8843	0.0486
A2	Attention	8/16	0.141	3.640	6,099	1,263.4	0.8987	0.0418
A3	Attn.+MLP	8/16	0.595	3.979	8,398	917.6	0.9250	0.0392
A4	**Attn.+MLP**	**32/64**	**2.336**	**4.422**	**8,765**	**879.2**	**0.9384**	**0.0401**
A5	Attn.+MLP	32/16	2.336	4.436	8,295	929.0	0.9211	0.0395
A6	Attn.+MLP	32/64	2.336	3.400	11,182	689.1	0.9168	0.0388

A1–A5 use LoRA: low-rank adapters with frozen base weights. A6 uses QLoRA, storing base weights in 4-bit NF4 with double quantization. Attention = q, k, v, o; MLP = gate, up, down. Memory includes the base model and adapter; adapter scale is α/r.

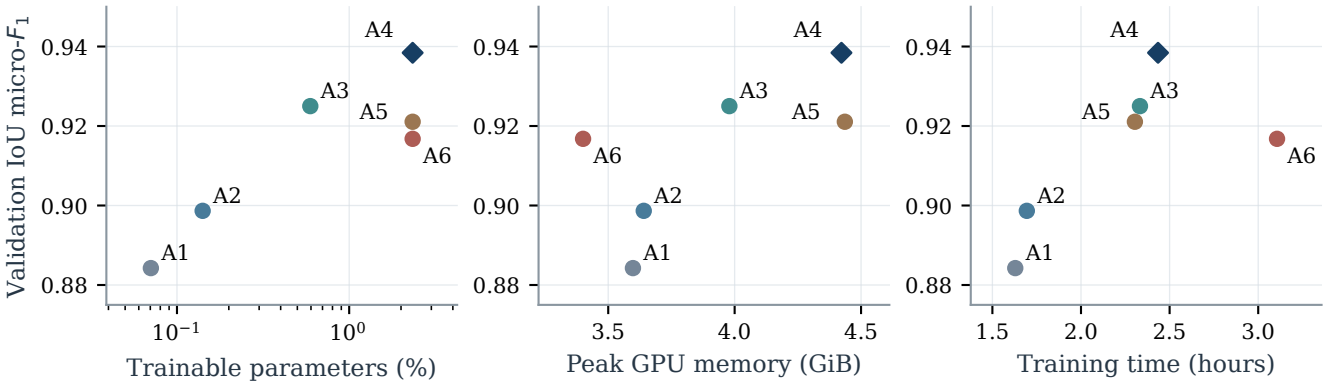


Figure 1: Validation accuracy versus measured costs. A4 is the diamond. The parameter axis is logarithmic. Points are individual runs.

Placement, rank, and scaling. Extending adapters to all attention projections (A1→A2) changes F_1 by +1.44 percentage points (pp); also adapting the MLP (A2→A3) adds +2.63 pp. At fixed $\alpha = 16$, raising rank from 8 to 32 (A3→A5) changes F_1 by -0.39 pp. At rank 32, raising α to 64 (A5→A4) adds +1.74 pp: a higher rank alone does not ensure a better sanitizer [2]. Compared with A4, QLoRA A6 uses 23.11% less memory, but loses 2.16 pp in F_1 and takes 27.58% longer to train [3].

Efficiency and choice. A1 gives the highest F_1 per million trainable parameters (0.812); A6 leads per GiB (0.270), and A1 per training hour (0.543). These ratios compare the measured runs, not gains guaranteed by extra resources. The final A3→A4 gain (+1.34 pp) is the smallest along A1→A2→A3→A4, suggesting diminishing returns. A4 alone is within .01 of the best validation F_1; the leakage, overmasking and memory tie-breaks therefore do not change its selection.

3 Part B: sanitization quality and generalization

Part B compares A4 with hand-written regular-expression (regex) rules and Presidio's default recognizers on the same 2,000 in-distribution (ID) and 1,000 out-of-distribution (OOD) examples. ID data follow the training distribution; OOD data test performance on a different distribution. A4 chooses the most likely next token (greedy decoding), with batch 8 and a 512-token output limit. Neither baseline is adjusted using the OOD set.

Table 2: Sanitization results on both test sets. Higher precision, recall and F_1 are better; lower leakage and overmasking are better. Rates are on $[0, 1]$, rounded to four decimals; $< .0001$ denotes a smaller positive rate.

	ID: 2,000 examples			OOD: 1,000 examples		
Metric	A4	Regex	Presidio	A4	Regex	Presidio
Exact entity matching						
Micro precision	0.9088	0.7502	0.2102	0.2320	0.5586	0.0945
Micro recall	0.9541	0.2012	0.1313	0.1593	0.1012	0.0591
Micro F_1	0.9309	0.3173	0.1616	0.1889	0.1714	0.0727
Macro precision	0.9084	0.1703	0.1099	0.2015	0.0697	0.0519
Macro recall	0.9417	0.1540	0.1054	0.1365	0.0784	0.0468
Macro F_1	0.9237	0.1583	0.1074	0.1229	0.0707	0.0397
IoU $\geq$ .5 entity matching						
Micro precision	0.9117	0.7762	0.2164	0.2405	0.5586	0.1078
Micro recall	0.9572	0.2082	0.1352	0.1652	0.1012	0.0674
Micro F_1	0.9339	0.3283	0.1664	0.1958	0.1714	0.0829
Macro precision	0.9126	0.1770	0.1140	0.2132	0.0697	0.0588
Macro recall	0.9462	0.1604	0.1095	0.1432	0.0784	0.0536
Macro F_1	0.9280	0.1649	0.1114	0.1305	0.0707	0.0465
Leak rate $\downarrow$	0.0381	0.7448	0.4738	0.3248	0.8185	0.4143
Overmask rate $\downarrow$	0.0026	0.0030	0.0178	0.0007	< 0.0001	0.0104
Well-formed rate $\uparrow$	0.9895	1.0000	1.0000	0.9910	1.0000	1.0000

Table 3: A4 on the ten most frequent ID reference labels. Support is the number of annotated entities. P = precision; R = recall. Label names follow the supplied dataset.

		Exact			IoU $\geq$ .5		
Class	Support	P	R	F_1	P	R	F_1
TIME	945	0.7603	0.9598	0.8485	0.7611	0.9608	0.8494
USERNAME	760	0.9681	0.9579	0.9630	0.9681	0.9579	0.9630
SOCIALNUMBER	668	0.9862	0.9596	0.9727	0.9877	0.9611	0.9742
EMAIL	662	0.9834	0.9864	0.9849	0.9880	0.9909	0.9894
IDCARD	651	0.9433	0.9447	0.9440	0.9433	0.9447	0.9440
DRIVERLICENSE	643	0.9581	0.9238	0.9406	0.9581	0.9238	0.9406
LASTNAME1	623	0.7779	0.8941	0.8320	0.7779	0.8941	0.8320
BOD	553	0.9624	0.9729	0.9676	0.9678	0.9783	0.9730
PASSPORT	552	0.8463	0.9475	0.8940	0.8528	0.9547	0.9009
SEX	529	0.9357	0.9905	0.9624	0.9357	0.9905	0.9624

3.1 Scoring and error analysis

Unchanged text aligns each predicted mask with its source span (exclusive end offset). Reference spans are used only for scoring. Repeated text, adjacent masks and rewrites can make alignment ambiguous. Exact matches require the same boundaries and label; overlap matches require the same label and intersection over union (IoU) $\geq .5$. Matching is one-to-one and maximizes total IoU, the overlap length divided by the union length.

Precision is the fraction of predictions matched; recall is the fraction of reference entities found; F_1 is their harmonic mean. Micro scores pool entities; macro scores average supported classes. Leakage is the fraction of reference values still present after case and whitespace normalization. Overmasking is the fraction of non-PII characters removed or replaced. A wrong label can therefore lower F_1 even when the value is removed.

Table 4: The 586 ID reference entities missed by A4 under exact matching.

Miss type	Count
Incorrect label	315
Verbatim value retained	122
Unrecoverable/incorrect replacement	110
Boundary mismatch	39

Main finding. IoU F_1 drops by 73.80 pp on OOD data; leakage rises from 3.81% to 32.48%. For labels seen in training, OOD recall is 84.39% (519/615); for 47 unseen labels, it is 0.68% (18/2,636). Ignoring label names raises OOD IoU F_1 to 0.7644. Thus, label mismatch explains much of the detection-score drop, but the higher leakage still shows a loss of protection. Presidio's broader labels also affect strict matching.

4 Part C: extraction and the privacy–utility trade-off

V-base is Qwen2.5-0.5B, a *base* language model with no local fine-tuning. V-raw is trained on the raw corpus; V-san is trained on its A4-sanitized version. Both use LoRA on all attention and MLP projections ($r = 32, \alpha = 64$), seed 42, four epochs, and the optimizer and batch settings in Section 1. Only the training text differs. Sanitization keeps all 8,007 rows, their repetition counts and metadata; planted secrets receive no special treatment.

Utility is measured on 500 held-out documents (65,458 tokens) using perplexity (PPL). PPL is calculated by summing negative log-likelihood (NLL) over all scored tokens, dividing by the token count, and exponentiating. Lower PPL means better next-token prediction. Documents are weighted by their token counts rather than averaging document-level perplexities.

Table 5: Utility and memorization across the three models. Discoverable memorization tests 3,520 training-document prefixes. Higher exposure means planted secrets rank more favorably. Both attacks are detailed in Section 5.

		Discoverable memorization		
Model	Held-out PPL	Exact rate	Edit similarity	Mean exposure (bits)
V-base	15.2105	0.0000	0.2703	1.5350
V-raw	8.3316	0.0011	0.3907	3.8680
V-san	13.3986	0.0009	0.3259	0.8423

4.1 Untargeted extraction and membership scoring

Each model generates 5,000 samples starting only from a beginning-of-sequence token. Sampling uses temperature 1, top-$k = 40$, top-$p = 1$, and at most 256 new tokens. Duplicate and empty outputs are excluded from ranking. Four scores rank the remaining outputs from smallest to largest, preserving their original order for ties [4]. Here $L_v(x)$ is mean token negative log-likelihood (NLL) under model v:

$$S_{\text{ppl}} = e^{L_v(x)}, \quad S_{\text{ref}} = e^{L_v(x) - L_{\text{base}}(x)}, \quad S_{\text{lower}} = e^{L_v(x) - L_v(\text{lower}(x))}, \quad S_{\text{zlib}} = \frac{L_v(x)}{H_{\text{zlib}}(x)}. \tag{1}$$

Here $H_{\text{zlib}} = 8\,|\text{zlib.compress}(x)|/|x|$ uses default compression and UTF-8 byte lengths. The scores measure likelihood directly or relative to the base model, lowercased text, or compressibility. Precision@100 (P@100) is the fraction of the 100 highest-ranked outputs containing an exact 50-token training-text match. V-san is checked against its *sanitized* training text: reproducing a masked passage is not necessarily disclosing PII.

Table 6: Untargeted extraction results. *Raw docs* counts distinct original documents with a 50-token match anywhere in the sample pool, not only in the top 100 outputs.

		P@100 by ranking score				
Model	Unique outputs	PPL	Reference	Lowercase	Zlib	Raw docs
V-base	5,000	0.0000	0.0000	0.0000	0.0000	0
V-raw	4,991	0.2900	0.0000	0.0100	0.1200	19
V-san	4,848	0.3500	0.1600	0.0300	0.5100	22

Against the original raw text instead, V-san's P@100 is .1400, .0800, .0000 and .2300 for PPL, reference, lowercase and zlib, respectively.

4.2 Targeted prompting

The supplied prompts target 200 training-corpus members and 200 non-member controls. Each receives one greedy continuation and ten sampled continuations, with at most 64 new tokens and the sampling settings above. Only the continuation is checked for the reference value. *Any10* is the fraction of prompts with a hit in at least one of ten draws, not the success rate per draw.

Table 7: Targeted value recovery. Hits from the unmodified base model or on non-member controls show that a match need not result from training on the local corpus.

	Members ($n = 200$)		Controls ($n = 200$)	
Model	Greedy	Any10	Greedy	Any10
V-base	0.0050	0.0150	0.0300	0.0850
V-raw	0.0750	0.0750	0.0450	0.0950
V-san	0.0100	0.0150	0.0400	0.0550

What changes after sanitization? Member Any10 recovery falls from 7.5% to 1.5% (15 to 3 of 200 prompts), and mean exposure falls from 3.8680 to 0.8423 bits. However, PPL rises from 8.3316 to 13.3986: a 60.82% increase, meaning worse prediction on held-out text. Sanitization reduces these extraction measures at a utility cost; it does not stop text memorization. V-san's higher sanitized-text P@100 may reflect repeated mask templates; this explanation is untested. Four prompts already contain the reference value (one member, three controls). They remain in these rates but are excluded from the traced examples.

5 Repetition effects and traced sanitizer failures

5.1 Discoverable memorization

This attack asks whether a model can reproduce training text when given its beginning [5]. For each distinct non-canary document, the first 50 content tokens prompt a greedy prediction of the next 50. Of 5,000 documents, 1,480 have fewer than 100 tokens and are excluded, leaving 3,520. V-base reproduces the continuation exactly for 0 documents, V-raw for 4, and V-san for 3. All three are compared with the original raw text. Edit similarity uses the token-level `SequenceMatcher` ratio: 1 means an identical continuation, and 0 means no matched tokens.

Table 8: Discoverable exact-match rates by the supplied document repetition count. Unequal group sizes and very few recoveries limit trend claims.

Repetitions	Eligible documents	V-base	V-raw	V-san
1	3,379	0.0000	0.0003	0.0003
2	49	0.0000	0.0000	0.0000
8	45	0.0000	0.0222	0.0444
32	47	0.0000	0.0426	0.0000

5.2 Canary exposure

Canaries are artificial secrets inserted into training text at known repetition counts. For each of 20 six-digit secrets, compare the true value with 9,999 unique alternatives [6]. Candidates are ranked by summed NLL over secret tokens only, excluding suffix and end-of-sequence tokens. Leading zeros are preserved; tied candidates are counted ahead of the true secret. If its rank among 10,000 candidates is r_s, the estimated rank in the million-value space is $\widehat{r}$. Exposure is the reduction in log-rank relative to that space:

$$\widehat{r} = 10^6 \frac{r_s}{10000}, \qquad \widehat{\text{exposure}} = \log_2(10^6) - \log_2 \widehat{r} = \log_2 \frac{10000}{r_s}. \tag{2}$$

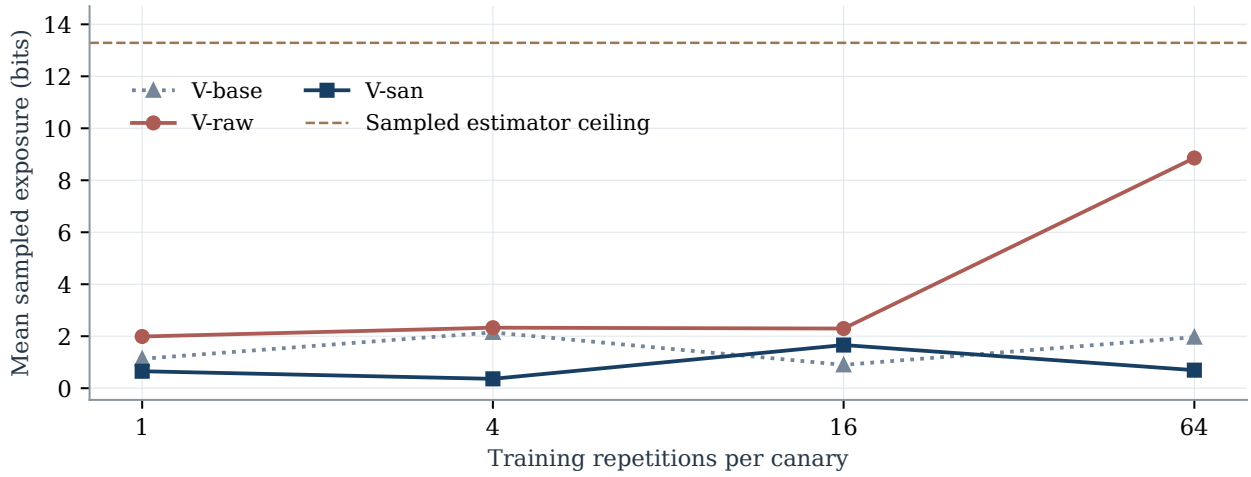


Figure 2: Mean canary exposure at each repetition count (five different secrets per group). Higher exposure means the model ranks the secret more favorably. The 10,000-candidate estimate cannot distinguish exposure above 13.2877 bits.

V-raw rises sharply at 64 repetitions, whereas V-san has lower mean exposure overall. Compared with V-raw, V-san lowers exposure for 15/20 secrets but raises it for 5/20. The raw curve is not strictly monotone. Five canaries per group and sampled ranks support a descriptive comparison, not a statistical or worst-case guarantee.

5.3 Sanitizer failure: a redacted trace

Document `doc_00051`; prompt `member_046`. A one-character `SEX` value, `[REDACTED]`, occurs in a training document repeated twice. A4 leaves it unmasked, and V-san later produces a continuation containing it. The value is absent from the attack prompt. This links a missed mask to a later output match, although it does not establish that training caused the match.

Model	Greedy	Any10
V-base	No	Yes
V-raw	No	Yes
V-san	Yes	Yes

Caution. The unmodified base model also produces a sampled match. A single character may occur inside an unrelated word, making this weak evidence of training-data leakage. Only one example met the tracing criteria; no second qualifying example was found.

6 Discussion and limitations

6.1 Training and evaluation settings

Each sanitizer completes 1,250 optimizer updates; each trained victim completes 2,004 updates over 32,028 row visits. Training uses FP16 computation with FP32 adapters, gradient checkpointing, and a 1,024-token sequence limit. Oversize inputs are rejected rather than truncated. If mixed-precision arithmetic overflows, the same accumulation group is retried with a smaller loss scale and the same random state; no training rows are skipped.

Experiments run on T4 GPUs. Concurrent jobs use separate GPUs but share host CPU and input/output resources, so wall times also reflect contention. All three reported perplexities use likelihood batch 64 on the same 65,458 held-out tokens. Attack sampling uses seed 42 with explicitly fixed decoding settings; no model-specific generation defaults are used.

6.2 Execution times

Table 9: Evaluation wall times, including time spent before and after a resumed run. The last column shows the limit under a per-invocation interpretation; aggregate totals are discussed below.

Evaluation	Minutes	Limit (min)
A validation (A1–A3)	48.45, 60.82, 82.14	90 each
A validation (A4–A6)	84.10, 85.48, 118.54	90 each
B inference (ID; OOD)	89.12; 18.09	90 each
C attacks (base; raw; san)	55.04; 109.23; 90.31	180 each

A6 validation (118.54 min) exceeded the 90-minute cap. Combined B ID+OOD (107.21 min) and three-model attacks (254.59 min) exceed 90/180 min if the caps are aggregate. Durations include resumed work; the aggregate scope of the limits has not been clarified.

6.3 Next steps

The next improvement would be a sanitizer that recognizes a broader range of PII labels and contexts. The large OOD drop and frequent label errors suggest combining broader training-label coverage with context-sensitive rules. Model choices should use validation data, measuring leakage and overmasking alongside exact and overlap F_1: removing more text is not automatically better.

Repetition should be addressed separately through a controlled deduplication comparison, followed by the same extraction attacks and held-out utility evaluation. This would distinguish the effects of removing repeated text from those of masking PII. Where a formal guarantee is needed, differentially private fine-tuning could also be tested, with an explicit privacy budget and a measured utility cost. These extensions were not implemented in this assignment.

6.4 Limitations

All configurations use one seed, so the results do not measure variation across training runs. Finite attack samples cannot show that unrecovered data are safe, and sampled exposure has an upper resolution limit. Span alignment and substring matching can introduce scoring errors, especially for short values. A 50-token match establishes text overlap, not necessarily disclosure of personal information. The supplied bundle lacks the official full-corpus PII index, so corpus-wide PII recovery is not reported. Only one qualifying trace was found, and some evaluations exceeded the specified time limits.

Conclusion. The selected sanitizer performs well on familiar labels and reduces targeted recovery and average canary exposure. However, its sharp OOD decline, remaining output matches and higher held-out perplexity show that sanitization is a partial defense, with both coverage limitations and a measurable utility cost.

Source code. github.com/adeemharis/LLM_Privacy. Tool-use details and acknowledgments are in `README.md`.

References

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[3] T. Dettmers et al., “QLoRA: Efficient Finetuning of Quantized LLMs,” NeurIPS, 2023. arXiv:2305.14314.
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